

Medical Statistics - Answers lab 5

Introduction

This answer page gives one complete set of answers for lab 5. The same numerical results should be obtained when the analyses are carried out in R or SPSS.

The analyses use the Cleveland heart disease dataset and address the following research questions:

- Does maximum heart rate achieved differ across chest pain types?
- Is the presence of heart disease associated with sex?
- Is the presence of heart disease associated with fasting blood sugar?

Exercise 1: Download and open the dataset

The dataset contains 303 patients and 14 variables.

```
library(haven)
library(knitr)
library(tableone)
library(car)
library(dunn.test)

heart_data <- as.data.frame(read_sav("datasets/heart_disease_cleveland.sav"))

# Convert labelled categorical variables to factors.
labelled_vars <- vapply(heart_data, haven::is_labelled, logical(1))
heart_data[labelled_vars] <- lapply(heart_data[labelled_vars], haven::as_factor)

dim(heart_data)
```

```
[1] 303 14
```

```
names(heart_data)
```

```
[1] "age"      "sex"      "cp"      "trestbps" "chol"     "fbs"
[7] "restecg"  "thalach"  "exang"    "oldpeak"  "slope"    "ca"
[13] "thal"     "target"
```

The variables `sex`, `cp`, `fbs`, `restecg`, `exang`, `slope`, `thal`, and `target` are categorical. The variable `ca` is also best treated as categorical for the baseline table, because it records the number of major vessels from 0 to 3.

```
heart_data$ca <- factor(heart_data$ca)
```

Exercise 2: Create a baseline characteristics table

```
continuous_vars <- c("age", "trestbps", "chol", "thalach", "oldpeak")
categorical_vars <- c("sex", "cp", "fbs", "restecg", "exang", "slope", "ca", "thal")
all_vars <- c(continuous_vars, categorical_vars)

table1 <- CreateTableOne(
  vars = all_vars,
  data = heart_data,
  factorVars = categorical_vars
)

kable(
  print(
    table1,
    nonnormal = "oldpeak",
    printToggle = FALSE
  )
)
```

	Overall
n	303
age (mean (SD))	54.44 (9.04)
trestbps (mean (SD))	131.69 (17.60)
chol (mean (SD))	246.69 (51.78)
thalach (mean (SD))	149.61 (22.88)

	Overall
oldpeak (median [IQR])	0.80 [0.00, 1.60]
sex = Male (%)	206 (68.0)
cp (%)	
Typical angina	23 (7.6)
Atypical angina	50 (16.5)
Non-anginal pain	86 (28.4)
Asymptomatic	144 (47.5)
fbs = True (%)	45 (14.9)
restecg (%)	
Normal	151 (49.8)
ST-T wave abnormality	4 (1.3)
Left ventricular hypertrophy	148 (48.8)
exang = Yes (%)	99 (32.7)
slope (%)	
Upsloping	142 (46.9)
Flat	140 (46.2)
Downsloping	21 (6.9)
ca (%)	
0	176 (58.9)
1	65 (21.7)
2	38 (12.7)
3	20 (6.7)
thal (%)	
Normal	166 (54.8)
Fixed defect	20 (6.6)
Reversible defect	117 (38.6)

A suitable baseline table gives one summary format per variable. In the table above, most continuous variables are summarized using mean and standard deviation, `oldpeak` is summarized using median and interquartile range because it is clearly skewed, and categorical variables are summarized using counts and percentages.

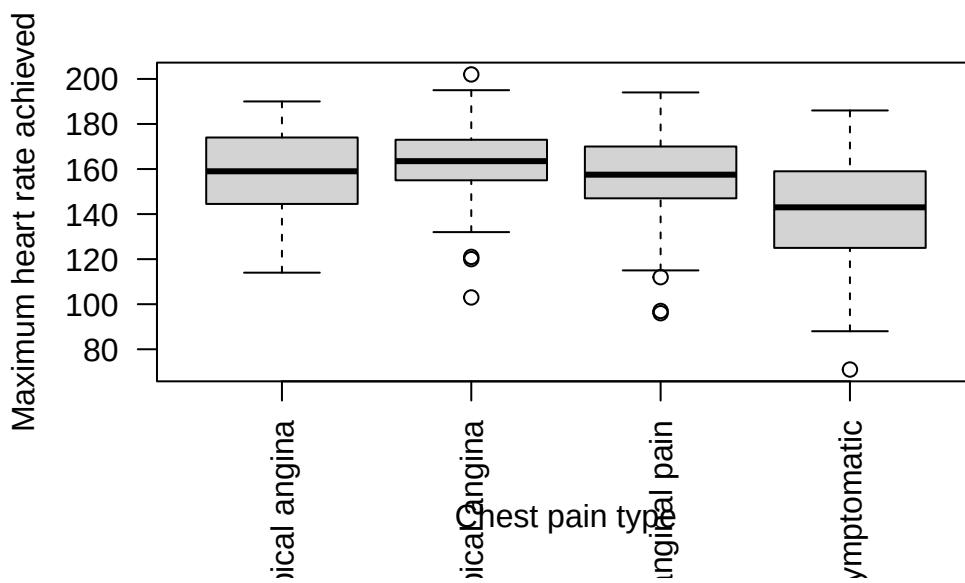
Exercise 3: Analyze maximum heart rate across chest pain types

Research question: *Does maximum heart rate achieved differ across chest pain types?*

The outcome is `thalach`, and the grouping variable is `cp`.

Descriptive results

```
boxplot(  
  thalach ~ cp,  
  data = heart_data,  
  xlab = "Chest pain type",  
  ylab = "Maximum heart rate achieved",  
  las = 2  
)
```



```
for (group in levels(heart_data$cp)) {  
  group_data <- heart_data$thalach[heart_data$cp == group]  
  n <- length(group_data)  
  mean_thalach <- mean(group_data)  
  se_thalach <- sd(group_data) / sqrt(n)  
  ci_thalach <- mean_thalach + c(-1, 1) * qt(0.975, df = n - 1) * se_thalach  
  
  cat(  
    group,  
    ": n =", n,  
    ", mean =", round(mean_thalach, 1),  
    ", 95% CI = [", round(ci_thalach[1], 1), ", ", round(ci_thalach[2], 1), "]\n",  
    file = "results.txt", append = TRUE)
```

```

    sep = ""
  )
}

```

Typical angina: n =23, mean =156, 95% CI = [146.9, 165]
 Atypical angina: n =50, mean =162.4, 95% CI = [157, 167.8]
 Non-anginal pain: n =86, mean =155.4, 95% CI = [151.3, 159.5]
 Asymptomatic: n =144, mean =140.7, 95% CI = [136.9, 144.5]

The mean maximum heart rate was lowest in the asymptomatic chest pain group.

One-way ANOVA

```

anova_model <- aov(thalach ~ cp, data = heart_data)
summary(anova_model)

```

	Df	Sum Sq	Mean Sq	F value	Pr(>F)
cp	3	23503	7834	17.41	1.91e-10 ***
Residuals	299	134523	450		

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

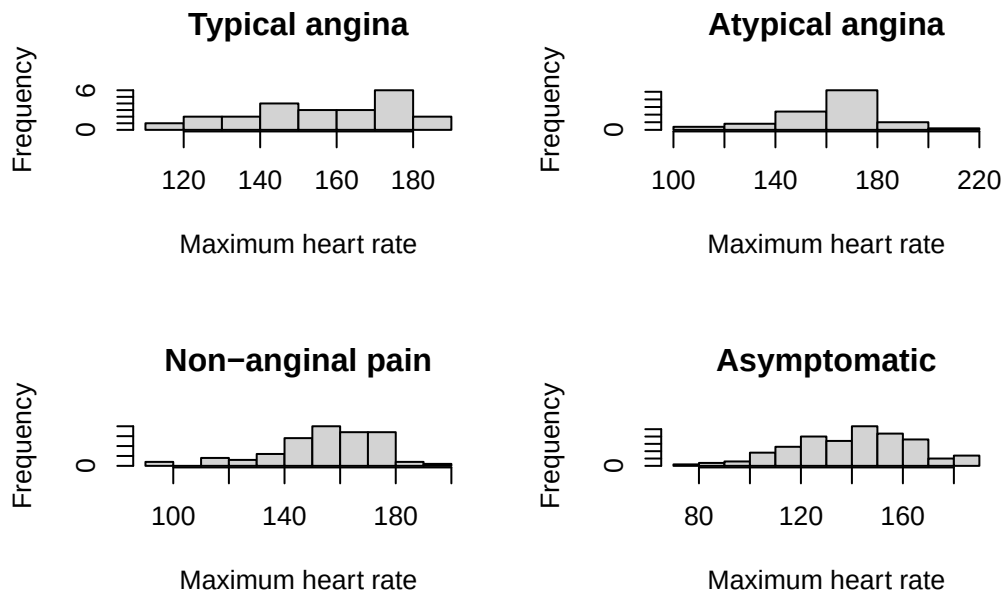
The one-way ANOVA is statistically significant, $F(3, 299) = 17.41$, $p < 0.001$. This indicates that maximum heart rate achieved differs across chest pain types.

Assumption checks

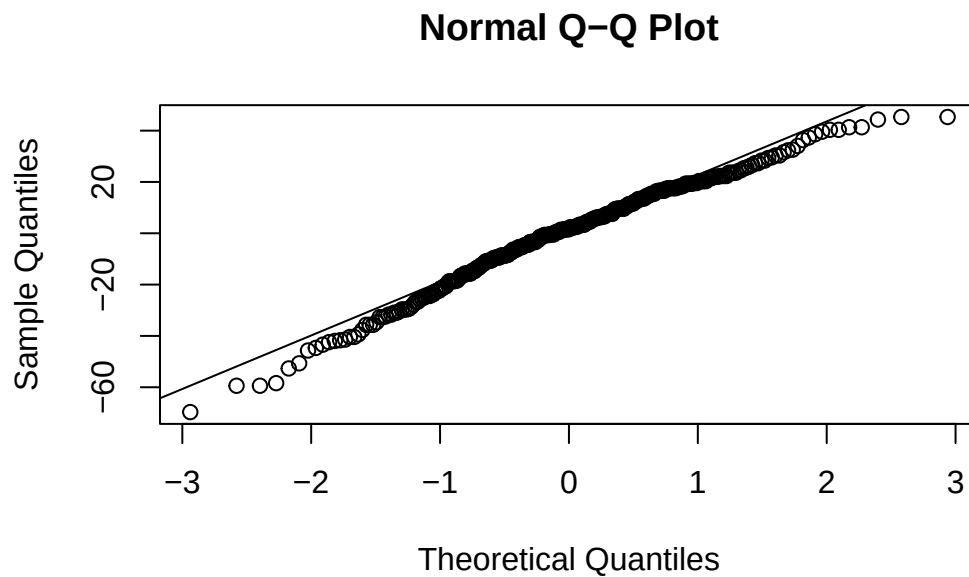
```

par(mfrow = c(2, 2))
hist(heart_data$thalach[heart_data$cp == "Typical angina"],
     main = "Typical angina", xlab = "Maximum heart rate")
hist(heart_data$thalach[heart_data$cp == "Atypical angina"],
     main = "Atypical angina", xlab = "Maximum heart rate")
hist(heart_data$thalach[heart_data$cp == "Non-anginal pain"],
     main = "Non-anginal pain", xlab = "Maximum heart rate")
hist(heart_data$thalach[heart_data$cp == "Asymptomatic"],
     main = "Asymptomatic", xlab = "Maximum heart rate")

```



```
par(mfrow = c(1, 1))
qqnorm(residuals(anova_model))
qqline(residuals(anova_model))
```



```
tapply(heart_data$thalach, heart_data$cp, sd)
```

Typical angina	Atypical angina	Non-anginal pain	Asymptomatic
20.89906	18.98882	19.31679	22.98210

```
leveneTest(thalach ~ cp, data = heart_data)
```

Levene's Test for Homogeneity of Variance (center = median)

	Df	F value	Pr(>F)
group	3	2.4694	0.06209
	299		

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

The histograms and Q-Q plot can be used to assess whether the distributions are roughly compatible with ANOVA. There is no need to base this decision on a formal test of normality; with these sample sizes such tests are often not helpful for deciding which analysis to use. The group standard deviations are of a similar order of magnitude, and Levene's test does not show evidence against the equal variances assumption.

Post-hoc comparisons

```
pairwise.t.test(
  heart_data$thalach,
  heart_data$cp,
  p.adjust.method = "bonferroni",
  pool.sd = TRUE
)
```

Pairwise comparisons using t tests with pooled SD

data: heart_data\$thalach and heart_data\$cp

	Typical angina	Atypical angina	Non-anginal pain
Atypical angina	1.0000	-	-
Non-anginal pain	1.0000	0.3840	-
Asymptomatic	0.0089	8.8e-09	3.7e-06

P value adjustment method: bonferroni

Using Bonferroni-adjusted pairwise t-tests, the asymptomatic chest pain group has a significantly lower maximum heart rate than the typical angina group, the atypical angina group, and the non-anginal pain group. The other pairwise comparisons are not statistically significant after Bonferroni adjustment.

Kruskal-Wallis sensitivity analysis

```
kruskal.test(thalach ~ cp, data = heart_data)
```

Kruskal-Wallis rank sum test

data: thalach by cp

Kruskal-Wallis chi-squared = 47.315, df = 3, p-value = 2.979e-10

```
dunn.test(
  heart_data$thalach,
  heart_data$cp,
  method = "bonferroni",
  altp = TRUE
)
```

Kruskal-Wallis rank sum test

data: x and group

Kruskal-Wallis chi-squared = 47.3151, df = 3, p-value = 0

Comparison of x by group (Bonferroni)

Col Mean-			
Row Mean	Asymptom	Atypical	Non-angi
-----+	-----		
Atypical	-5.970501		
	0.0000*		
Non-angi	-4.836917	1.804218	
	0.0000*	0.4272	
Typical	-2.983404	1.230885	-0.045777
	0.0171*	1.0000	1.0000


```
alpha = 0.05
Reject Ho if p <= alpha
```

The Kruskal-Wallis test gives the same substantive conclusion, $\chi^2(3) = 47.32$, $p < 0.001$. The Bonferroni-adjusted Dunn's test again shows that the asymptomatic group differs from the other three chest pain groups.

Exercise 4: Analyze heart disease by sex

Research question: *Is the presence of heart disease associated with sex?*

Note

For the 2x2 tables in this lab, `chisq.test(..., correct = FALSE)` is used to report the ordinary Pearson chi-square test. This corresponds to the **Pearson Chi-Square** row in SPSS. If `correct = FALSE` is omitted, R applies Yates' continuity correction by default for 2x2 tables; SPSS reports this separately as **Continuity Correction**.

```
sex_table <- table(heart_data$sex, heart_data$target)
sex_table
```

	Heart Disease	No Heart Disease
Female	25	72
Male	114	92

```
round(prop.table(sex_table, margin = 1) * 100, 1)
```

	Heart Disease	No Heart Disease
Female	25.8	74.2
Male	55.3	44.7

The row percentages show that the prevalence of heart disease was higher in males than in females.

The expected cell counts are all larger than 5, so the chi-square test is appropriate.

```
sex_chisq <- chisq.test(sex_table, correct = FALSE)
sex_chisq$expected
```

	Heart Disease	No Heart Disease
Female	44.49835	52.50165
Male	94.50165	111.49835

```
sex_chisq
```

Pearson's Chi-squared test

```
data: sex_table
X-squared = 23.218, df = 1, p-value = 1.446e-06
```

There is strong evidence of an association between sex and heart disease, $\chi^2(1) = 23.22$, $p < 0.001$. The prevalence of heart disease was higher in males (55.3%) than in females (25.8%).

Exercise 5: Analyze heart disease by fasting blood sugar

Research question: *Is the presence of heart disease associated with fasting blood sugar?*

```
fbs_table <- table(heart_data$fbs, heart_data$target)
fbs_table
```

	Heart Disease	No Heart Disease
False	117	141
True	22	23

```
round(prop.table(fbs_table, margin = 1) * 100, 1)
```

	Heart Disease	No Heart Disease
False	45.3	54.7
True	48.9	51.1

The row percentages show similar heart disease prevalence in the two fasting blood sugar groups.

The expected cell counts are all larger than 5, so the chi-square test is appropriate.

```
fbs_chisq <- chisq.test(fbs_table, correct = FALSE)
fbs_chisq$expected
```

	Heart Disease	No Heart Disease
False	118.35644	139.64356
True	20.64356	24.35644

```
fbs_chisq
```

Pearson's Chi-squared test

```
data: fbs_table
X-squared = 0.19339, df = 1, p-value = 0.6601
```

There is no evidence of an association between fasting blood sugar and heart disease, $\chi^2(1) = 0.19$, $p = 0.660$. The prevalence of heart disease was similar for patients with fasting blood sugar at or below 120 mg/dl (45.3%) and patients with fasting blood sugar above 120 mg/dl (48.9%).